Designing and Manufacturing a Protective and Interactive Helmet for Children

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Abstract

The main objective of this project is to design and manufacture a protective and interactive helmet tailored for children that is cost-effective and easily producible using 3D printing, aiming to enhance child safety by providing physical protection and immediate awareness of impact location in the event of a fall. SolidWorks was used for 3D modeling and Ultimaker Cura for slicing, with PLA filament selected as the manufacturing material and the helmet produced using an Ender 3 V2 printer. The electronic system utilizes an Arduino Nano microcontroller, five pushbuttons, and five LEDs to create the impact indication feature by illuminating the LED corresponding to the area of impact. The final prototype is lightweight, adjustable for different head sizes, and practical for everyday use, offering an effective solution to help reduce risks associated with childhood falls.

Introduction

In today's rapidly evolving era, it is difficult to predict when, where, or how accidents may occur, as their consequences can lead to injuries of varying severity or even death. According to the World Health Organization, a fall is defined as an event that causes a person to come to rest inadvertently on the ground or a lower level, and globally, falls are the second leading cause of unintentional injury deaths, with over 37 million people requiring medical attention annually and millions of life years lost due to resulting disabilities. Existing protective solutions in the market are limited in accessibility and effectiveness, highlighting the need for a reliable and affordable alternative. Therefore, this project focuses on designing and manufacturing a protective and interactive helmet for children that provides physical protection during falls and integrates an interactive system to help parents immediately identify the impact location and respond appropriately.

Project Objectives

The main objective of this project is to design and manufacture a protective and interactive helmet for children to ensure their safety during various activities. The helmet is designed to be durable and comfortable, providing effective protection without compromising ease of use. The project also aims to reduce production costs to make the product more affordable and accessible to a wider range of families.

Problem Statement

Helmets play a vital role in protecting children from head injuries caused by falls, which can lead to skull fractures, concussions, or serious complications. According to the World Health Organization, fall-related injuries are expected to increase by 100% by 2030 if preventive measures are not implemented, especially during activities such as biking, skating, and recreational play. Therefore, protective solutions are needed to reduce injury severity and help parents respond quickly. The proposed interactive helmet provides immediate visual indication of the impact location, enabling timely assessment and action. A community questionnaire involving 214 parents supported this need, as shown in Figure 1 and Figure 2, which demonstrate that dizziness and excessive sleepiness were the most commonly reported post-fall symptoms (29%), along with other symptoms such as head swelling, hallucinations, momentary amnesia, swallowing the tongue, and vomiting.

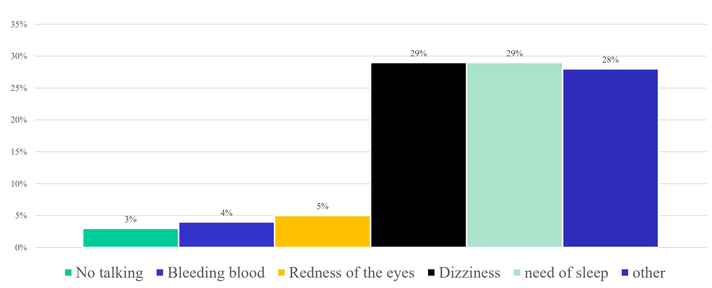

Figure 1. Fall Symptoms

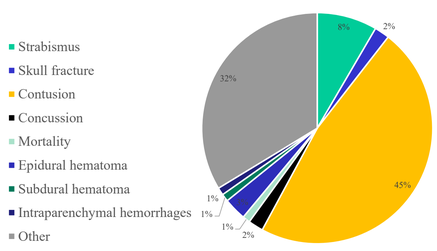

Figure 2. Head Injury Results.

Methodology

Phase 1: Design and Modeling

Based on community survey requirements, the helmet was designed using SolidWorks to ensure proper head coverage for children. The design considered head size range, ventilation, weight distribution, protection of vulnerable areas (front, sides, and back), and integration points for the impact indication system.

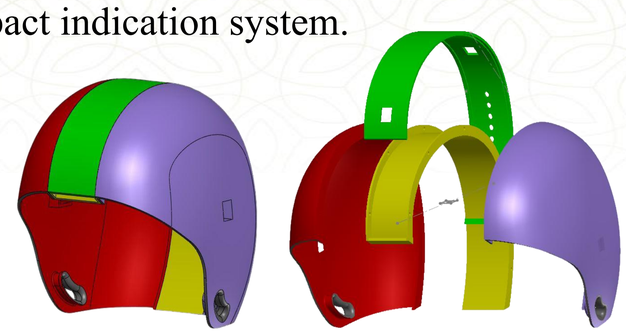

Figure 3. Helmet 3D Model

Phase 2: Printing

The finalized 3D helmet model was printed using an Ender 3 V2 FDM printer. PLA filament was selected due to its lightweight properties, biocompatibility, and cost-effectiveness, making it suitable for wearable and child-safe applications.

Phase 3: Electronics Circuit Implementation

An Arduino Nano was programmed using the Arduino IDE to connect five pushbuttons placed at key impact zones with corresponding LEDs. The system detects impact direction and visually indicates the affected area, powered by a lithium-ion battery.

Phase 4: Assembly and Testing

Electronic components, including the Arduino Nano, LEDs, pushbuttons, and battery, were installed in designated helmet compartments. Components were centrally positioned to maintain balance and comfort, secured using magnetic attachments, and the system was thoroughly tested for functionality and reliability.

Phase 5:Final Project

The final prototype integrates a 3D-printed protective helmet with an embedded interactive impact indication system. External LEDs provide clear visual feedback, while the lightweight, child-friendly design and rechargeable battery ensure comfort, portability, and user acceptance.

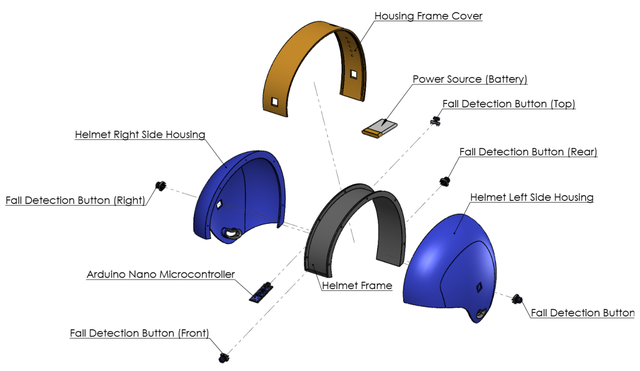

Figure 4. 3D Exploded Helmet Layout.

Figure 5. Final Helmet Prototype

Conclusion

This project resulted in a functional protective helmet designed based on real community needs and safety concerns for children. The helmet provides head protection and an interactive signaling system that helps parents quickly identify the impact location after a fall and supports better decision-making, particularly during early childhood when head injuries pose higher risks. Using 3D printing and electronic components made the design affordable, comfortable, and child-friendly, contributing to safer play environments and greater parental reassurance.

References

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